

Karish Grover

✉ karishg@cs.cmu.edu



<https://karish-grover.github.io>



[karishgrover](#)

✉ @karishgrover



[karish-grover](#)



[Google Scholar](#)

Education

- Carnegie Mellon University**
Ph.D., Machine Learning

◇ Advisors: [Prof. Christos Faloutsos](#) and [Prof. Geoffrey J. Gordon](#).
◇ [Amazon AI Ph.D. Fellowship](#) (a.y. 2025–2027)
◇ [Meta AIM Ph.D. Fellowship](#) (a.y. 2025–2027)

Pittsburgh, PA
2023 – 2028
- Indraprastha Institute of Information Technology, Delhi**
B.Tech., Computer Science and Artificial Intelligence | GPA: 9.2/10.0

◇ Best [Undergraduate Thesis Award](#).
◇ [Chanakya Undergraduate Fellowship](#) (a.y. 2022–2023)

Delhi, India
2019 – 2023

Refereed Publications

First Author

- [A1]** Feedback Control for Multi-Objective Graph Self-Supervision
K. Grover, T. Vasiloudis, H. Xie, S. Lu, X. Song, and C. Faloutsos
43rd International Conference on Machine Learning

[ICML 26]
- [A2]** Text Has Curvature
K. Grover, H. Zeng, Y. Xia, C. Faloutsos, and G. J. Gordon
43rd International Conference on Machine Learning

[ICML 26]
- [A3]** CurvGAD: Leveraging Curvature for Enhanced Graph Anomaly Detection
K. Grover, G. J. Gordon, and C. Faloutsos
42nd International Conference on Machine Learning

[ICML 25]
- [A4]** Spectro-Riemannian Graph Neural Networks
K. Grover, H. Yu, X. Song, Q. Zhu, H. Xie, V. N. Ioannidis, and C. Faloutsos
13th International Conference on Learning Representations

[ICLR 25]
- [A5]** Discourse-Aware Hyperbolic Fourier Co-Attention for Social-Text Classification
K. Grover, S. M. P. Angara, M. S. Akhtar, and T. Chakraborty
36th International Conference on Neural Information Processing Systems (Oral Paper)

[NeurIPS 22]

Other Publications

- [B1]** DAG-MoE: From Simple Mixture to Structural Aggregation in Mixture-of-Experts
J. Feng, H. Zeng, **K. Grover**, R. Qiu, Y. Xia, Q. Zhang, Q. Wang, R. Chen, D. Fu, J. Liu, Z. Zhao, X. Fan, B. Zhang, and Y. Chen
43rd International Conference on Machine Learning

[ICML 26]
- [B2]** Dependency Aware Incident Linking in Large Cloud Systems
S. Ghosh, **K. Grover**, J. Wong, C. Bansal, R. Namineni, M. Verma, and S. Rajmohan
ACM Web Conference 2024

[WWW 24]
- [B3]** Multi-Relational Graph Transformer for Automatic Short Answer Grading
R. Agarwal, V. Khurana*, **K. Grover***, M. Mohania, and V. Goyal
Conference of the North American Chapter of the Association for Computational Linguistics

[NAACL 22]

Patents

[C1] Dependency Aware Incident Linking in Large Cloud Systems

S. Ghosh, J. Wong, C. Bansal, R. Namineni, M. Verma, S. Rajmohan, and **K. Grover**

US Patent App. 18/520,931



[US Patent 24]

Work Experience

■ **Meta**, Visiting Researcher

Pittsburgh, PA

Advised by: **Dr. Yinglong Xia** and **Dr. Hanqing Zeng**

Sep 2025 – Present

◇ *Team*: Meta AI – Modern Recommendation Systems (MRS).

◇ *Description*: Graph ML and geometric representation learning for recommender systems; Mixed-curvature (geometric) generative recommender systems; Curvature-based language foundation models.

■ **Amazon**, Applied Scientist Intern

Santa Clara, CA

Advised by: **Dr. Theodore Vasiloudis** and **Dr. Xiang Song**

May 2025 – Aug 2025

◇ *Team*: Amazon Web Services (AWS) Deep Engine Science — Graph ML (AGML).

◇ *Description*: Multi-task self-supervised learning and control-theoretic scheduling on graphs; We introduce Control-G, the *first* control-theoretic approach to multi-objective graph SSL, which uses a task difficulty-aware Proportional-Integral-Derivative (PID) scheduler to decide which task to train next.

■ **Amazon**, Applied Scientist Intern

Santa Clara, CA

Advised by: **Dr. Xiang Song**, **Dr. Vassilis Ioannidis** and **Dr. Qi Zhu**

May 2024 – Aug 2024

◇ *Team*: Amazon Web Services (AWS) Deep Engine Science — Graph ML (AGML).

◇ *Project [A2]*: Geometric and spectral graph representation learning; We propose Spectro-Riemannian Graph Neural Networks (CUSP) - a mixed-curvature spectral graph neural network that learns spectral filters to optimize node embeddings in products of constant curvature manifolds.

■ **Microsoft**, Research Intern

Bengaluru, India

Advised by: **Dr. Xuchao Zhang** and **Dr. Supriyo Ghosh**

Jan 2023 – Jul 2023

◇ *Team*: Microsoft Research × M365 Research AIOps.

◇ *Project [C1], [B1]*: AIOps and incident management for large-scale cloud services; We propose a dependency-aware incident linking (DiLink) framework which leverages both textual and service dependency graph information to improve the accuracy and coverage of incident links.

■ **LinkedIn**, Applied Research Intern

Bengaluru, India

Manager: **S. M. Phaneendra Angara**

May 2022 – Aug 2022

◇ *Team*: LinkedIn AI — Misinformation & Virality.

◇ *Project [A3]*: Geometric representation learning for misinformation and virality; hyperbolic kernel annular attention with RL-based propagation-subgraph selection for explainable fake-news detection on social platforms.

Awards and Achievements

Oct 2025 ■ **Amazon AI Ph.D. Fellowship** — honored to receive the Amazon fellowship (a.y. 2025-2027) supporting my Ph.D. research at CMU. [

Sept 2025 ■ **Meta AIM Ph.D. Fellowship** — received the Meta fellowship (a.y. 2025-2026) to support my Ph.D. research, as part of the **Meta AI Mentorship program**. [

Oct 2023 ■ **Best Undergraduate Thesis Award**, IIIT-Delhi — university-wide award (gold medal) for UG thesis on geometric deep learning for graphs and language [A3]. [

Awards and Achievements (continued)

- Sept 2022  **Kartikeya Gupta Memorial Scholarship**, IIIT-Delhi — single university-wide awardee for outstanding all-round academic and co-curricular performance. []
- Aug 2022  **Dean's Award for Academic Excellence**, IIIT-Delhi — recognition for top-tier academic performance for three consecutive years (*a.y.* 2021-2023). []
- Nov 2021  **Chanakya UG Fellowship**, national cohort (20 awardees) — Received fellowship (*a.y.* 2021-2022) to support my undergraduate research at IIITD. Awarded by **iHub-Anubhuti**. []
- Jul 2021  **First Position, HAH@IberLEF 2021 (SEPLN) shared task** — winning system for humorous tweet classification at **HAHA subtask** at IberLEF conference. []
- Sep 2019  **Atal Tinkering Grant**, Intel & NITI Aayog — \$20,000 grant to establish a Tinkering Lab; built an assistive navigation robot for visually impaired users. []